claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_736be611-fb5\agent-
ada44152b85badf28.jsonl`
- SHA-256: `0dddc6cce0739b2e7ac09079593b649106f4c060f2b2238028dfcdcf5a3cd6d0`
- Source modified: `2026-06-16T15:14:30+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_736be611-fb5`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T15:12:42.309Z)

```
You are a small-data evaluation-rigor expert. The owner argues strict tIoU 0.5 over-penalizes
because the SERVICE WINDOW makes rally starts intrinsically ~1-1.5s fuzzy (even Gemini's median
|?start| is 1.46s), and proposes a +-2s tolerance. Critique the current metric set (tIoU-F1,
F1@k, merge/split, mIoU) for THIS task. Design the task-faithful metric: a +-tau tolerance
match, boundary-error-in-seconds split start/end, while KEEPING an over-segmentation guard so a
relaxed tolerance cannot hide merges. What tau, and how does it relate to inter-annotator
agreement? At n=6-7 what makes a delta trustworthy (CI + sign-test)? Cite numbers.


REAL-FOOTAGE VALIDATION DATA (6 Kushagra/Yash June-12 clips, 1080p proxies, fully-LOCAL no-AI
WASB
windowing vs Gemini. Gemini = strong REFERENCE, NOT ground truth ? these 6 have no golden labels
EXCEPT mahadevpura-2 which the owner hand-labeled, see GROUND-TRUTH below).

Rally COUNT (Gemini | local baseline | local W1 cap=12 | local W1+W2 mc=1):
  GX010128: 39 | 15 | 78 | 75
  GX010129: 27 | 2  | 50 | 47
  GX020125: 11 | 6  | 18 | 15
  mahadevpura-0: 40 | 19 | 62 | 43
  mahadevpura-1: 9 | 1 | 2 | 2
  mahadevpura-2: 39 | 5 | 40 | 40
  AGGREGATE: Gemini 165 | baseline 48 (mean window 62.9s, 27 merge-groups, mIoU 0.23) |
             W1 250 (mean 10.8s, 10 merge-groups, mIoU 0.39) |
             W1+W2 222 (mean 11.5s, 10 merge-groups, mIoU 0.39, 6 gemini-only "misses")
  Per-video mean window (W1): GX010128 7.8s, GX010129 9.3s, GX020125 14.1s, mahadevpura-0 10.4s,
             mahadevpura-1 86.9s (the continuously-active blob W1 cannot cap ? no inactivity
gap),
             mahadevpura-2 13.8s. Gemini means ~5.7-7.2s.

GROUND-TRUTH (mahadevpura-2, 40 owner-labeled rallies, mean 5.7s, verified frame-aligned):
  F1@0.5 / F1@0.25 / mean-IoU:  Gemini 0.557/0.835/0.55 ; local W1+W2 0.150/0.500/0.31 ;
  local W1+W2+hard-cap 0.296/0.722/0.50 (best local). baseline 0.000/0.000/0.03.
  BOUNDARY ERROR (matched rallies, seconds): Gemini |?start| med 1.46 |?end| med 1.19 ;
  local-hardcap |?start| med 1.74 (~= Gemini!) |?end| med 3.12 (>> Gemini).
  => local's rally-START detection is ~at par with Gemini; the gap is the rally-END.
  IoU<->time: for a 5.7s rally, tIoU 0.5 tolerates only ~+-1.9s shift (the ~1-1.5s service
window
  nearly exhausts it -> tIoU 0.5 over-penalizes; owner wants a +-2s tolerance metric).

PRIOR GOLDEN DELTAS (n=6, then n=7 incl. mahadevpura-2): baseline F1 0.300 -> W1 0.439 (+0.139,
6/0)
  -> W1+W2 0.457 (+0.019, 6/0); hard-cap golden-neutral on n=6 (+0.006 n.s.) but became BEST on
n=7
  (continuously-active clips). All flags default-OFF, promoted only on measured LOVO lift +
```

sign-test.

WINDOWING KNOBS: a frame is "active" iff shuttle visible AND per-second-normalized velocity > velocity_thresh(0.02); active frames within merge_gap(2.0s) coalesce; min_window(2.0s). W1 = max_window_duration cap(12s) split at the largest internal inactivity gap. W1.5 hard-cap = midpoint-chop when no gap. W2 = net-crossings gate (drop windows with <1 net crossing).

For raw per-video detail you MAY Read C:/Users/avidu/Projects/badminton-highlight-indexer/output/local_vs_gemini_6.json . Treat the DATA block as authoritative for the numbers. Constraints: local pipeline must stay MIT/Apache/BSD-only, single local GPU, ~6-7 golden videos now (~16 by the weekend), Gemini usable at inference only (never as a training-data source).

2. assistant (2026-06-16T15:12:46.032Z)

I'll analyze the data and design a task-faithful metric. Let me start by reading the raw per-video detail to ground my critique in the actual numbers.

3. user (2026-06-16T15:12:47.665Z)


```
[
  {
    "name": "GX010128",
    "vid": "db84f3e65318",
    "csv": "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\.claude\\worktrees\\w2-safeguard\\output\\wasb\\GX010128__db84f3e65318_wasb.csv",
    "n_gem": 39,
    "play_gem": 228.0,
    "mean_gem": 5.846153846153846,
    "variants": {
      "baseline": {
        "n": 15,
        "play": 711.4614614614616,
        "mean_dur": 47.430764097430774,
        "max_dur": 133.2499165832499,
        "count_ratio": 0.38461538461538464,
        "matched": 4,
        "local_only": 1,
        "gemini_only": 0,
        "splits": 0,
        "merges": 10,
        "mean_iou": 0.3564105364413525,
        "gem_rallies_in_merges": 35
      },
      "W1": {
        "n": 78,
        "play": 605.1051051051056,
        "mean_dur": 7.757757757757765,
        "max_dur": 16.616616616616625,
        "count_ratio": 2.0,
        "matched": 35,
        "local_only": 34,
        "gemini_only": 0,
        "splits": 4,
        "merges": 0,
        "mean_iou": 0.5294639963484731,
        "gem_rallies_in_merges": 0
      },
      "W1+W2": {
        "n": 75,
        "play": 594.494494494495,
        "mean_dur": 7.926593259926601,
        "max_dur": 16.616616616616625,
        "count_ratio": 1.9230769230769231,
        "matched": 35,
```

```
          "local_only": 31,
          "gemini_only": 0,
          "splits": 4,
          "merges": 0,
          "mean_iou": 0.5294639963484731,
          "gem_rallies_in_merges": 0
        }
      },
      "status": "ok"
    },
    {
      "name": "GX010129",
      "vid": "99a3e73afde3",
      "csv": "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\.claude\\worktrees\\w2-safeguard\\output\\wasb\\GX010129__99a3e73afde3_wasb.csv",
      "n_gem": 27,
      "play_gem": 195.6,
      "mean_gem": 7.2444444444444445,
      "variants": {
        "baseline": {
          "n": 2,
          "play": 534.4177510844178,
          "mean_dur": 267.2088755422089,
          "max_dur": 522.6893560226894,
          "count_ratio": 0.07407407407407407,
          "matched": 0,
          "local_only": 1,
          "gemini_only": 0,
          "splits": 0,
          "merges": 1,
          "mean_iou": 0.0,
          "gem_rallies_in_merges": 27
        },
        "W1": {
          "n": 50,
          "play": 463.7137137137137,
          "mean_dur": 9.274274274274273,
          "max_dur": 27.84451117784451,
          "count_ratio": 1.8518518518518519,
          "matched": 23,
          "local_only": 18,
          "gemini_only": 0,
          "splits": 4,
          "merges": 0,
          "mean_iou": 0.4819655005521238,
          "gem_rallies_in_merges": 0
        },
        "W1+W2": {
          "n": 47,
          "play": 447.0303636970303,
          "mean_dur": 9.511284333979368,
          "max_dur": 27.84451117784451,
          "count_ratio": 1.7407407407407407,
          "matched": 22,
          "local_only": 16,
          "gemini_only": 1,
          "splits": 4,
          "merges": 0,
          "mean_iou": 0.48514996335815863,
          "gem_rallies_in_merges": 0
        }
      },
      "status": "ok"
    },
    {
```

```
      "name": "GX020125",
      "vid": "407fa7e7e9f1",
      "csv": "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\.claude\\worktrees\\local-noai\\output\\wasb\\GX020125__407fa7e7e9f1_wasb.csv",
      "n_gem": 11,
      "play_gem": 71.0,
      "mean_gem": 6.454545454545454,
      "variants": {
        "baseline": {
          "n": 6,
          "play": 280.6806806806807,
          "mean_dur": 46.78011344678012,
          "max_dur": 82.66599933266599,
          "count_ratio": 0.5454545454545454,
          "matched": 1,
          "local_only": 2,
          "gemini_only": 0,
          "splits": 0,
          "merges": 3,
          "mean_iou": 0.40270862800565765,
          "gem_rallies_in_merges": 10
        },
        "W1": {
          "n": 18,
          "play": 254.03737070403744,
          "mean_dur": 14.113187261335414,
          "max_dur": 40.05672339005673,
          "count_ratio": 1.6363636363636365,
          "matched": 9,
          "local_only": 8,
          "gemini_only": 0,
          "splits": 0,
          "merges": 1,
          "mean_iou": 0.4329964346464835,
          "gem_rallies_in_merges": 2
        },
        "W1+W2": {
          "n": 15,
          "play": 246.5131798465132,
          "mean_dur": 16.434211989767547,
          "max_dur": 40.05672339005673,
          "count_ratio": 1.3636363636363635,
          "matched": 9,
          "local_only": 5,
          "gemini_only": 0,
          "splits": 0,
          "merges": 1,
          "mean_iou": 0.4329964346464835,
          "gem_rallies_in_merges": 2
        }
      },
      "status": "ok"
    },
    {
      "name": "mahadevpura-0",
      "vid": "6460c2fc7647",
      "csv": "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\.claude\\worktrees\\w2-safeguard\\output\\wasb\\mahadevpura-0__6460c2fc7647_wasb.csv",
      "n_gem": 40,
      "play_gem": 262.0,
      "mean_gem": 6.55,
      "variants": {
        "baseline": {
          "n": 19,
          "play": 711.5615615615616,
```

```
          "mean_dur": 37.45060850324008,
          "max_dur": 134.13413413413414,
          "count_ratio": 0.475,
          "matched": 6,
          "local_only": 4,
          "gemini_only": 0,
          "splits": 0,
          "merges": 9,
          "mean_iou": 0.3446682872977475,
          "gem_rallies_in_merges": 34
        },
        "W1": {
          "n": 62,
          "play": 646.1127794461132,
          "mean_dur": 10.421173862034085,
          "max_dur": 47.66433099766434,
          "count_ratio": 1.55,
          "matched": 34,
          "local_only": 22,
          "gemini_only": 0,
          "splits": 2,
          "merges": 2,
          "mean_iou": 0.4928109196809172,
          "gem_rallies_in_merges": 4
        },
        "W1+W2": {
          "n": 43,
          "play": 549.5161828495164,
          "mean_dur": 12.779446112779452,
          "max_dur": 47.66433099766434,
          "count_ratio": 1.075,
          "matched": 30,
          "local_only": 9,
          "gemini_only": 5,
          "splits": 1,
          "merges": 2,
          "mean_iou": 0.4837616679798798,
          "gem_rallies_in_merges": 4
        }
      },
      "status": "ok"
    },
    {
      "name": "mahadevpura-1",
      "vid": "38bef75d78cb",
      "csv": "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\.claude\\worktrees\\local-noai\\output\\wasb\\mahadevpura-1__38bef75d78cb_wasb.csv",
      "n_gem": 9,
      "play_gem": 51.2,
      "mean_gem": 5.688888888888889,
      "variants": {
        "baseline": {
          "n": 1,
          "play": 174.2742742742743,
          "mean_dur": 174.2742742742743,
          "max_dur": 174.2742742742743,
          "count_ratio": 0.1111111111111111,
          "matched": 0,
          "local_only": 0,
          "gemini_only": 0,
          "splits": 0,
          "merges": 1,
          "mean_iou": 0.0,
          "gem_rallies_in_merges": 9
        },
```

```
          "W1": {
            "n": 2,
            "play": 173.79045712379047,
            "mean_dur": 86.89522856189524,
            "max_dur": 101.88521855188523,
            "count_ratio": 0.2222222222222222,
            "matched": 0,
            "local_only": 0,
            "gemini_only": 0,
            "splits": 0,
            "merges": 2,
            "mean_iou": 0.0,
            "gem_rallies_in_merges": 9
          },
          "W1+W2": {
            "n": 2,
            "play": 173.79045712379047,
            "mean_dur": 86.89522856189524,
            "max_dur": 101.88521855188523,
            "count_ratio": 0.2222222222222222,
            "matched": 0,
            "local_only": 0,
            "gemini_only": 0,
            "splits": 0,
            "merges": 2,
            "mean_iou": 0.0,
            "gem_rallies_in_merges": 9
          }
        },
        "status": "ok"
      },
      {
        "name": "mahadevpura-2",
        "vid": "038e0e0915a2",
        "csv": "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\.claude\\worktrees\\local-noai\\output\\wasb\\mahadevpura-2__038e0e0915a2_wasb.csv",
        "n_gem": 39,
        "play_gem": 266.0,
        "mean_gem": 6.82051282051282,
        "variants": {
          "baseline": {
            "n": 5,
            "play": 605.9059059059059,
            "mean_dur": 121.18118118118119,
            "max_dur": 297.03036369703034,
            "count_ratio": 0.1282051282051282,
            "matched": 2,
            "local_only": 0,
            "gemini_only": 0,
            "splits": 0,
            "merges": 3,
            "mean_iou": 0.2945165777074605,
            "gem_rallies_in_merges": 37
          },
          "W1": {
            "n": 40,
            "play": 551.0844177510842,
            "mean_dur": 13.777110443777104,
            "max_dur": 43.3266599933267,
            "count_ratio": 1.0256410256410255,
            "matched": 12,
            "local_only": 4,
            "gemini_only": 0,
            "splits": 4,
            "merges": 5,
```

```
          "mean_iou": 0.4204774017180335,
          "gem_rallies_in_merges": 11
        },
        "W1+W2": {
          "n": 40,
          "play": 551.0844177510842,
          "mean_dur": 13.777110443777104,
          "max_dur": 43.3266599933267,
          "count_ratio": 1.0256410256410255,
          "matched": 12,
          "local_only": 4,
          "gemini_only": 0,
          "splits": 4,
          "merges": 5,
          "mean_iou": 0.4204774017180335,
          "gem_rallies_in_merges": 11
        }
      },
      "status": "ok"
    }
  ]
```


4. assistant (2026-06-16T15:13:01.959Z)

I have everything I need. The DATA block plus the per-video JSON are sufficient ? this is an evaluation-design question, not a web-research one. Let me reason it through and return the structured analysis.

A few key derivations I'll anchor on, computed from the data:

IoU?time conversion (for a rally of duration D, a pure translation by ? with equal-length prediction gives IoU = (D??)/(D+?)):
- Solving IoU = 0.5 ? ? = D/3. For D = 5.7s (owner GT mean): ? = 1.9s. For D = 6.82s (Gemini mean on mahadevpura-2): ? = 2.27s.
- Solving IoU = 0.25 ? ? = 0.6·D = 3.42s at D=5.7s.
- So tIoU 0.5 ? a ±1.9s tolerance and tIoU 0.25 ? a ±3.4s tolerance ? *for translation only*. For a symmetric shrink/grow it's even tighter.

The diagnostic split (from GROUND-TRUTH): local |?start| median 1.74s ? Gemini's 1.46s, but local |?end| median 3.12s vs Gemini's 1.19s. A single tIoU number blends these; it cannot show that start is at-par and end is the entire gap. This is the central indictment of the current metric.

Over-segmentation is the live failure mode, not merging (W1+W2 count_ratio: GX010128 1.92, GX010129 1.74, GX020125 1.36, mahadevpura-0 1.075, mahadevpura-2 1.03; aggregate 222 local vs 165 Gemini = 1.35). A pure tolerance match with greedy 1:1 assignment would let these extra predictions hide for free ? that's exactly the trap to guard against.

Let me also verify the sign-test arithmetic for n=6 and n=7 before I commit it to recommendations.

5. user (2026-06-16T15:13:04.628Z)

```
n= 5
  k=0/5  two-sided p=1.0000
  k=1/5  two-sided p=1.0000
  k=2/5  two-sided p=1.0000
  k=3/5  two-sided p=1.0000
  k=4/5  two-sided p=0.3750
  k=5/5  two-sided p=0.0625
n= 6
  k=0/6  two-sided p=1.0000
  k=1/6  two-sided p=1.0000
  k=2/6  two-sided p=1.0000
  k=3/6  two-sided p=1.0000
```

```
  k=4/6  two-sided p=0.6875
  k=5/6  two-sided p=0.2188
  k=6/6  two-sided p=0.0312
n= 7
  k=0/7  two-sided p=1.0000
  k=1/7  two-sided p=1.0000
  k=2/7  two-sided p=1.0000
  k=3/7  two-sided p=1.0000
  k=4/7  two-sided p=1.0000
  k=5/7  two-sided p=0.4531
  k=6/7  two-sided p=0.1250
  k=7/7  two-sided p=0.0156
```

6. assistant (2026-06-16T15:13:10.102Z)

The sign-test arithmetic confirms the small-n reality: at n=6 you need a clean 6/0 sweep to even reach p=0.031 (one-sided ~0.016); at n=7 you need 7/0 (two-sided p=0.0156). This is decisive for what "trustworthy" can mean. The prior golden deltas being "6/0" is exactly why baseline?W1 (+0.139) was trustworthy while W1?W1+W2 (+0.019, also 6/0 but tiny effect) is direction-trustworthy but magnitude-negligible.

Let me also verify the IoU?time relationship and the over-segmentation precision penalty arithmetic that anchors my guard design.

7. user (2026-06-16T15:13:16.327Z)

```
D=5.70 tIoU=0.50 -> max translation delta=1.90 s
D=5.70 tIoU=0.25 -> max translation delta=3.42 s
D=6.82 tIoU=0.50 -> max translation delta=2.27 s
D=6.82 tIoU=0.25 -> max translation delta=4.09 s
D=5.69 tIoU=0.50 -> max translation delta=1.90 s
D=5.69 tIoU=0.25 -> max translation delta=3.41 s
---over-seg precision under tolerance matching---
npred=40 nref=40 matched=12 -> P=0.300 R=0.300
aggregate count ratio local/gem = 1.3454545454545455
```

8. assistant (2026-06-16T15:13:18.939Z)

All arithmetic confirmed. I have everything needed to deliver the critique and the metric design.

9. user (2026-06-16T15:14:16.556Z)

Structured output provided successfully

10. assistant (2026-06-16T15:14:30.167Z)

I've delivered the critique and metric design via StructuredOutput. Summary of the analysis:

The owner is right (arithmetically): tIoU 0.5 tolerates only ? = D/3 of pure boundary shift ? 1.90s at the GT mean rally (D=5.7s). The service-window start fuzz (Gemini median |?start|=1.46s) alone consumes ~0.77?1.0 of that budget before any end error, so tIoU 0.5 functionally is a sub-2s tolerance that the irreducible ambiguity nearly exhausts.

Core indictment of the current set: a single tIoU/mIoU blends start and end error, hiding the one fact that matters ? local START is already at Gemini parity (1.74s ? 1.46s) and the *entire* gap is rally-END (3.12s vs 1.19s). The metric can't direct the next fix.

The task-faithful metric:
- **Asymmetric tolerance match:** |?start| ? ?_start=2.0s AND |?end| ? ?_end=1.0?1.5s. Start gets 2× the inter-annotator noise floor (service window is fuzzy); end stays tight because rally-end is a sharp event and is exactly where local fails.
- **Over-segmentation guard (so a relaxed ? can't hide merges):** strict 1:1 global (Hungarian) assignment, report Precision-at-? alongside Recall-at-?. The 222/165 = 1.35 surplus then costs

precision directly and can't match for free.
- **Fragmentation/merge pair** as a hard no-regression gate (catches both W1 over-seg and the baseline mega-window pathology).
- **Boundary error reported as two separate signed+absolute distributions**, never folded into IoU.

Trustworthiness at n=6-7 (sign-test is binding, not CI): n=6 needs a clean 6/0 sweep for p=0.031 (5/1 ? p=0.219); n=7 needs 7/0 for p=0.016 (6/1 ? p=0.125). A bootstrap mean-F1 CI at this n spans ~±0.15?0.20 and will overlap zero ? report it as an honesty band, not a gate. This is why baseline?W1 (+0.139, 6/0) is real while W1?W1+W2 (+0.019, 6/0) is directionally-clean-but-negligible.

Relevant file: `C:\Users\avidu\Projects\badminton-highlight-indexer\output\local_vs_gemini_6.json` (per-video matched/merge/split/count_ratio detail) and the tracked doc to record ? + the IoU?time derivation: `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\GOLDEN_REGRESSION_FIXTURES.md`.